

Wireless Network Security Implementation Report

1. Introduction

Wireless network security protects systems and users from unauthorized access to the network and wireless infrastructure. For this implementation, a wireless network was configured using **WPA2 security**, with a **Wireless Intrusion Prevention System (WIPS)** used to identify and prevent unauthorized wireless access.

The implementation provides two layers of protection:

- **WPA2:** Protects wireless communications through authentication and encryption.
- **WIPS:** Monitors the wireless environment and responds to unauthorized devices or access points.

2. WPA2 Wireless Network Configuration

SSID: SOC-Lab-WiFi

Security: WPA2-Personal

Encryption: AES

Authentication: Pre-Shared Key (PSK)

Example configuration:

SSID: SOC-Lab-WiFi

Security Mode: WPA2-Personal

Encryption: AES

Authentication: PSK

Password: [Strong unique password]

Purpose

WPA2 was implemented to prevent unauthorized users from connecting to the wireless network and to protect wireless traffic from being transmitted without encryption.

A strong, unique wireless password was used rather than a default router password.

Security Benefit

WPA2 provides authentication before a device is permitted to join the network and encrypts wireless communications between authorized devices and the access point.

3. WIPS Implementation

Wireless Intrusion Prevention System

A **WIPS** was configured to monitor wireless activity for unauthorized devices and access points.

The WIPS monitors the wireless environment and identifies devices that are not authorized to participate in the SOC lab network.

Example WIPS Policy

WIPS Policy: Unauthorized Wireless Device Prevention

Authorized SSID:

SOC-Lab-WiFi

Authorized Access Point:

SOC-LAB-AP

Unauthorized Device:

Any unapproved wireless client or access point

Action:

Detect → Alert → Prevent/Block

Purpose

The WIPS provides an additional security layer beyond WPA2. While WPA2 protects the legitimate wireless network, WIPS monitors the surrounding wireless environment for suspicious or unauthorized wireless activity.

4. Example Detected Event

Unauthorized Access Point Detected

A simulated unauthorized wireless access point was introduced into the lab environment using an SSID similar to the legitimate network.

WIPS ALERT

Event: Unauthorized Access Point

SSID: SOC-Lab-WiFi

BSSID: AA:BB:CC:DD:EE:FF

Status: Unauthorized

Severity: High

Action: BLOCKED

The WIPS identified the unauthorized access point and initiated the configured prevention response.

Security Response

Unauthorized AP

↓

WIPS Detection

↓

Security Alert

↓

Device Identified

↓

WIPS Prevention



Unauthorized Access Blocked

This demonstrates how WIPS can help prevent attacks involving unauthorized wireless infrastructure.

5. WPA2 + WIPS Defense-in-Depth

The two controls provide complementary protection:

Security Control	Function
WPA2	Authenticates wireless clients and encrypts wireless traffic
WIPS	Monitors for unauthorized wireless devices/APs
WIPS Prevention	Blocks or disrupts unauthorized wireless access

WPA2 protects the **legitimate network**, while WIPS protects the **wireless environment surrounding the network**.

6. Relevance to the AI-Agentive SOC Project

Wireless security events can also become security alerts for the project's SOC triage engine. The project is designed to ingest structured security alerts and investigate them using multiple sources of evidence.

For example, a WIPS alert identifying an unauthorized access point could be submitted to the AI-agentive investigation workflow:

```
{
  "alert_type": "Unauthorized Wireless Access Point",
  "ssid": "SOC-Lab-WiFi",
  "bssid": "AA:BB:CC:DD:EE:FF",
  "severity": "High",
  "action": "Blocked"
}
```

The system could then correlate the event with available identity, SIEM, and endpoint information and produce an investigation report for the analyst.

7. Conclusion

The wireless security implementation establishes a layered defense using **WPA2 and WIPS**. WPA2 provides authentication and encryption for the authorized wireless network, while WIPS continuously monitors for unauthorized wireless devices and access points.

The simulated unauthorized access-point event demonstrates the prevention capability of the WIPS. When combined with the AI-Agentive SOC Triage Engine, wireless security alerts can become part of a broader security investigation workflow.

Overall, the implementation demonstrates how wireless authentication, encryption, monitoring, and prevention can work together to protect a security lab network and provide actionable events for SOC investigation.